Visual Biomechanics and Exercise Decisions

HN-FE02 | Final examination | 100 points | 120 minutes

Read actual force geometry, then connect exercise choices to nutrition and programming.

Name _____ Date _____

Answer all eight questions. Q1-4 carry 10 points each; Q5-8 carry 15 points each. Calculator permitted. Carry full precision until display. All cases are fictional. Use extra paper as needed; label each answer by question and part. No physical test is required.

Question 1 Read the geometry

HN-FE02-Q01 | 10 points

Use Figure V 1. It is abstract planar geometry with one external force; all numeric labels are in the image.

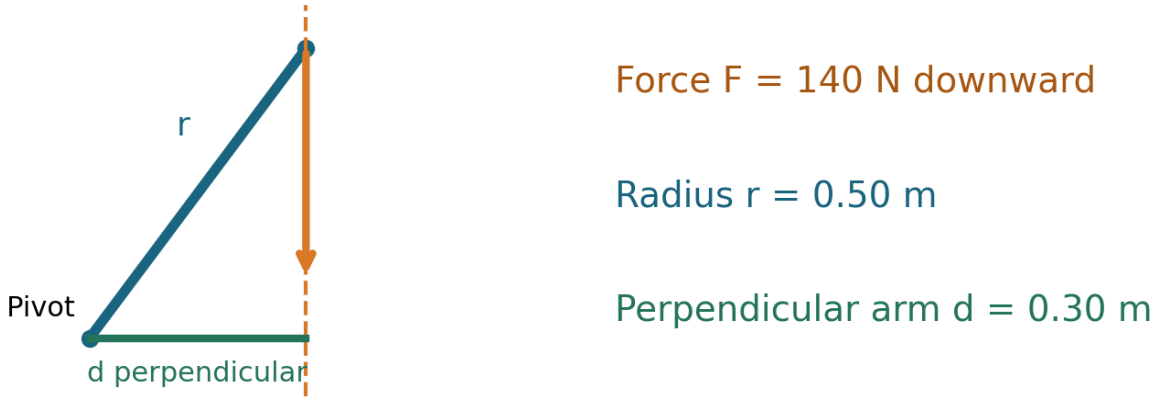


Figure V1 Original teaching geometry with assigned inputs.

a) [4 points] Identify force, radius, and perpendicular moment arm from the image.

b) [3 points] Calculate the force moment about the pivot.

c) [3 points] If only the perpendicular distance changes to 0.10 m, calculate the new moment and explain what is not established about internal tissue force.

Question 2 Two joint anatomy

HN-FE02-Q02 | 10 points

Compare hip flexion while keeping knee position otherwise comparable. Do not infer exact muscle lengths from an exercise name.

a) [4 points] Describe the directional effect on biarticular hamstrings and rectus femoris.

b) [3 points] Name the hamstring exception that does not cross the hip.

c) [3 points] Explain why replacing a knee curl with a hip thrust changes the primary task.

Visual Biomechanics and Exercise Decisions

Question 3 Compare profiles from a visual

HN-FE02-Q03 | 10 points

Figure V 2 supplies constant tension and three perpendicular moment arms for two abstract setups. Start, middle, end labels do not name a target muscle length.

Setup	Force N	Start d m	Middle d m	End d m
A	80	0.25	0.15	0.05
B	80	0.10	0.30	0.20

Figure V2 Original teaching geometry with assigned inputs.

a) [4 points] Calculate all three moments for each setup.

b) [3 points] Identify which setup has the greater moment at each position.

c) [3 points] Can the figure establish that the start is lengthened or that one setup produces more growth?

Question 4 Press and pull observations

HN-FE02-Q04 | 10 points

A coach pins the scapula down throughout overhead pressing, calls 60-degree elbow flare universally safe, and says wide-grip pulldown guarantees a wider back.

a) [4 points] Give a task-specific correction for the overhead scapular cue.

b) [3 points] Evaluate the universal elbow-angle claim.

c) [3 points] Distinguish scapular retraction from shoulder horizontal abduction and assess the grip claim.

Visual Biomechanics and Exercise Decisions

Question 5 Nutrition scenario

HN-FE02-Q05 | 15 points

VIS-N supplies 72 kg, 170 cm, 30 years, MSJ branch-161, multiplier 1.5, intake+5%, protein 1.8 and fat 0.8 g/kg BW. Use MSJ=10W+6.25H-5A+s and 4/9/4.

a) [5 points] Calculate RMR, TDEE scenario, and intake.

b) [5 points] Calculate protein, fat, and remaining carbohydrate.

c) [5 points] Explain why these precise outputs do not identify exact needs or regional muscle gain.

Question 6 Volume and frequency

HN-FE02-Q06 | 15 points

The table explicitly assigns contributions to one target. Each listed set is a physical set; the direct and indirect target categories do not duplicate rows here.

Day	Direct sets	Indirect sets
Mon	2	2
Wed	0	2
Fri	1	2

a) [5 points] Calculate direct-only, fractional(alpha 0.5), and total-contribution counts.

b) [5 points] Count direct exposure days and all contributing exposure days.

c) [5 points] Explain why frequency counts distinct days and why per-muscle counts need not sum to whole-program physical sets.

Visual Biomechanics and Exercise Decisions

Question 7 Progression and a conditional block

HN-FE02-Q07 | 15 points

Two distinct comparable exposures at 40 kg each contain three sets of 12 repetitions at 3 RIR. Range 8-12, minimum RIR 2, total increment 2.5 kg, no concern. Separately, a one-category six-week template begins at 7 direct sets.

a) [5 points] Apply the progression policy and calculate percentage increase.

b) [5 points] For offsets 0, 1, 2, 2, 3 and week 6=ceil(week 5/2), give all weekly counts and total.

c) [5 points] If the next exposure uses a shorter ROM, does it automatically validate same-task progress? Explain and state one review condition before adding sets.

Question 8 Substitution and scheduling

HN-FE02-Q08 | 15 points

An overhead cable triceps station is unavailable. An overhead dumbbell extension and a pressdown are available. Resistance is 17:00-18:00, proposed cycling 17:40-18:10, and sleep opportunity 00:00-07:00.

a) [5 points] Compare what the two substitutes preserve or change; avoid exact-profile or symptom-specific guarantees.

b) [5 points] Calculate overlap and sleep opportunity; propose a feasible scheduling change without promising full recovery.

c) [5 points] Evaluate the statement that standing calf raises must outperform seated training for every calf muscle.